Enrico Saccon

RESEARCH INTERESTS: LLM-ASSISTED ROBOTICS, TASK AND MOTION PLANNING, EMBODIED AI

Profile

PhD researcher in AI and robotics working on LLM-assisted planning, task and motion planning, and executable temporal plans for robotic systems. Experience with ROS/ROS2, simulation, optimization, logic programming, and research software development. Focused on building AI-enabled robotic systems that connect high-level reasoning, validation, and real-world deployment.

Skills

AI Planning	Multi-agent planning, task (temporal) planning, task and motion planning, multi-agent path finding, logic programming, probabilistic reasoning (MDPs).
Robotics	ROS(1,2), Gazebo, MoveIt, motion planning.
Optimization	OR-Tools, IBM CPLEX, constraint optimization, scheduling, dynamic programming.
LLMs and ML	Azure OpenAI SDK, Hugging Face Transformers, vLLM, PyTorch, TensorFlow, prompt-based knowledge extraction, prompt validation, reproducible experiments for LLM-assisted robotic planning systems.
Programming	Python, C, C++, Prolog, MATLAB, Bash, JavaScript.
Software Tools	Git, CUDA, Linux, Django, Node.js.

Work Experience

Mar 2025 – Nov 2025 **Research Internship**, *Cognitive Robotics Department – DLR*, Munich, Germany
Worked on integrating task and motion planning for the generation of executable temporal plans for multi-agent robotic systems. Focused on connecting symbolic task-level reasoning with imitation learning (KMP), optimization, and execution-oriented validation.
Topics: Task and Motion Planning (TAMP), Kernelized Movement Primitives (KMP), temporal planning, LLMs, optimization, robotic execution.

Sept 2019 – Dec 2019 **Computer Scientist**, *CreateNet – FBK*, Trento, Italy
Worked on cutting-edge technologies for control and optimization of agricultural irrigation in a deployed system.
Topics: C programming language, LoRaWAN infrastructure, electronic sensor and actuators.

Jan 2019 – Jul 2019 **High School Teacher**, *ITT Buonarroti-Pozzo*, Trento, Italy
Taught computer science to high school students:
1^{st} year: mainly problem solving skills; 2^{nd} year: basics of C programming.

Education

Nov 2022 – Current **PhD in Information Engineering and Computer Science**, *University of Trento*, Italy
Topics: Multi-Agent Planning, Temporal Planning, Industrial Robotics.
Goal: Develop a holistic system that through large language models and logic programming is able to plan and verify schedules for industrial robots and execute them, learning from experience.
GitHub: idra-lab/PLANTOR

Oct 2018 – Jul 2022 **Master's Degree in Computer Science**, *University of Trento*, Italy, Final mark: 109
Thesis title: "Comparison of Multi-Agent Path Finding Algorithms for an Industrial Scenario."
Thesis argument: managing a fleet of AGVs in a human populated environment.
Topics: AGV control, robotics principles, path and goal planning, fleet control.
Other strong acquired knowledge:
- Machine learning and deep learning (Tensorflow and PyTorch);
- Real time operating systems;

Sep 2015 – Oct 2018 **Bachelor's Degree in Computer Science**, *University of Trento*, Italy
"Implementation of GPU algorithms for robot path planning."
Topics: CUDA GPU programming, robot motion planning, comfort control.

Research Experience

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Trento, Italy | Open to relocate

Sep 2022 – Oct 2022 **Research Fellowship – "Predoc"**, *University of Trento*, Italy
Topics: Multi-Agent Path Finding, fleet management
Goal: Creation of a framework encompassing different MAPF algorithms for testing and scalability analysis

Dec 2020 – May 2021 **Research Project as Student**, *University of Trento*, Italy
- Research on **Dubins** curves for optimal control of vehicles;
- Implementation on **GPU** of dynamic programming for the multi-point Dubins problem;
- **Energetic analysis** of different solutions from embedded systems to server based ones.

Software and Research Artifacts

PLANTOR A framework for task and motion planning with LLMs and logic programming.
GitHub: idra-lab/PLANTOR

MDP Generator A framework for generating MDPs through logic programming and large language models.
GitHub: idra-lab/prolog_mdp

MPDP A framework to solve multi-point motion planning problems exploiting dynamic programming for Dubins and Reeds-Shepp. It also employs ML and DL models to improve performance.
GitHub: icosac/MPDP

MAOF A framework for testing and comparing different Multi-Agent Path Finding (MAPF) algorithms.
GitLab: chaff800/MAOF

Selected Publications

[6] M. Frego, E. Saccon, D. De Martini, and L. Palopoli, "Complexity Reduction of the Three-Point Dubins Problem (3PDP) Via Symmetry Exploitation for Machine Learning Purposes," *2026 International Conference on Robotics and Automation (ICRA)*, 2026.

[5] E. Saccon, D. De Martini, M. Saveriano, E. Lamon, L. Palopoli, and M. Roveri, "Automated Generation of MDPs Using Logic Programming and LLMs for Robotic Applications," *IEEE Robotics and Automation Letters*, vol. 11, no. 2, pp. 1770–1777, 2026. DOI: `10.1109/LRA.2025.3643276`.

[4] P. Pastorelli, S. Dagnino, E. Saccon, M. Frego, and L. Palopoli, "Fast shortest path polyline smoothing with G^1 continuity and bounded curvature," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, pp. 3182–3189, 2025. DOI: `10.1109/LRA.2025.3540531`.

[3] E. Saccon, A. Tikna, D. D. Martini, E. Lamon, L. Palopoli, and M. Roveri, *A temporal planning framework for multi-agent systems via llm-aided knowledge base management*, 2025. arXiv: `2502.19135 [cs.AI]`. [Online]. Available: `https://arxiv.org/abs/2502.19135`.

[2] E. Saccon, A. Tikna, D. De Martini, E. Lamon, L. Palopoli, and M. Roveri, "When prolog meets generative models: A new approach for managing knowledge and planning in robotic applications," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2024, pp. 17 065–17 071.

[1] E. Saccon, P. Bevilacqua, D. Fontanelli, M. Frego, L. Palopoli, and R. Passerone, "Robot Motion Planning: can GPUs be a Game Changer?" *2021 IEEE 45th Annual Computers, Software, and Applications Conference (COMPSAC)*, pp. 21–30, 2021. DOI: `10.1109/COMPSAC51774.2021.00015`.

Selected Conference Presentations

2026 ICRA: Presented work on automated MDP generation using logic programming and LLMs for robotic applications.

2025 IROS: Presented work on the three-point Dubins problem with machine learning models.

2024 ICRA & ICAPS: Presented work on generating Prolog-based knowledge management with LLMs and planning in robotics.

Teaching Experience

2023–2026 Teaching Assistant / Tutor, Robot Planning & its Applications. Lectures, laboratory exercises, and exams.

2022–2023 Teaching Assistant for Real Time Operating Systems and Programming 101.

Training, Leadership, and Communication

Training ICAPS 12th Summer School on task and motion planning, Banff, Canada, 2024. EUSCOTHA Winter School on control theory and applications, University of Trento, 2024.

Conference organization Local chair for NGEN-AI 2026, IDFC 2026, and QUANCOM 2026, supporting the local organization of international AI-related conferences hosted in Trento.

Collaboration Worked across academic, industrial, and international research environments, including University of Trento, FBK, and DLR. Participated in Hackathon Italia, Hackathon FBK, Hackathon Google, and Google Hash Code, with emphasis on rapid prototyping, algorithmic problem solving, and teamwork.

Awards and Grants

Jun 2024 **ICAPS24 DC Support**, Banff, Canada. As part of the DC, I was awarded a support of ~ 300\$ to register at the ICAPS24 conference.

Jul 2023 **SoCS23 Travel Grant**, Prague, Czech Republic. I was awarded a travel grant of 400 euros.

Languages

English Full professional knowledge

German Basic knowledge; completed A1a–A1b coursework

Italian Native